CASE STUDY:

E-COMMERCE SALES AND PROFITABILITY ANALYSIS

Author: Gopika

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1. Introduction

E-commerce businesses operate in highly competitive environments where data-driven decision-making is essential for sustaining growth and profitability. Every transaction generates valuable data that can be analyzed to understand customer purchasing patterns, product performance, and operational efficiency.

This case study focuses on analyzing an e-commerce dataset to uncover insights related to **sales performance, profitability, discount strategies, and regional trends**. The goal is to transform raw transactional data into meaningful insights that can support strategic business decisions.

2. Business Objective

The primary objective of this analysis is to **identify key factors influencing revenue and profitability** in an e-commerce business.

The analysis is designed to help stakeholders:

- Understand which product segments drive revenue
- Identify profitability gaps across product categories
- Evaluate the impact of discount strategies
- Compare regional business performance
- Support data-driven decision-making

3. Business Questions

To achieve the objective, the following analytical questions were defined:

- Which sub-categories generate the highest and lowest sales?
- Which sub-categories are the most and least profitable?
- How does discount impact profit?
- Which region generates the highest profit?

These questions form the foundation of the analysis and guide the data exploration process.

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4. Data Source

The dataset used in this project is the **Superstore dataset**, a widely used dataset for business analytics practice. It contains transactional-level data representing orders placed across different regions and product categories.

Key Fields in the Dataset

Column	Description
Category	High-level product category
Sub-Category	Detailed product classification
Region	Geographic sales region
Sales	Revenue generated from transactions
Profit	Profit earned from sales
Discount	Discount applied to the sale
Quantity	Number of units sold

This dataset provides sufficient information to analyze both **revenue and profitability drivers**.

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5. Tools Used

The following tools were used to complete this project:

- **Google Sheets**
  Used for initial data inspection, cleaning, and formatting.
- **MySQL**
  Used to perform structured data analysis through SQL queries, including aggregation and grouping.
- **Tableau Public**
  Used to create interactive dashboards and visualizations to communicate insights effectively.

6. Data Cleaning and Preparation

Before analysis, the dataset was reviewed and cleaned to ensure accuracy and usability.

Cleaning Steps Performed:

- Checked for missing or null values in key columns
- Verified numerical fields such as Sales, Profit, and Discount
- Standardized categorical fields (Category, Sub-Category, Region)
- Ensured consistency in data formatting
- Removed any duplicate or inconsistent entries (if present)

The dataset was then structured in a format suitable for SQL-based aggregation and analysis.

7. Data Analysis Approach

The analysis was conducted using SQL queries to extract meaningful insights from the dataset.

Key Analytical Techniques Used:

- **Aggregation (SUM, AVG)** to calculate totals and averages

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- **Grouping (GROUP BY)** to segment data by categories
- **Sorting (ORDER BY)** to identify top and bottom performers
- **Comparative analysis** across categories and regions

Analysis Performed:

- Total sales by sub-category
- Total profit by sub-category
- Average profit across discount levels
- Total profit by region

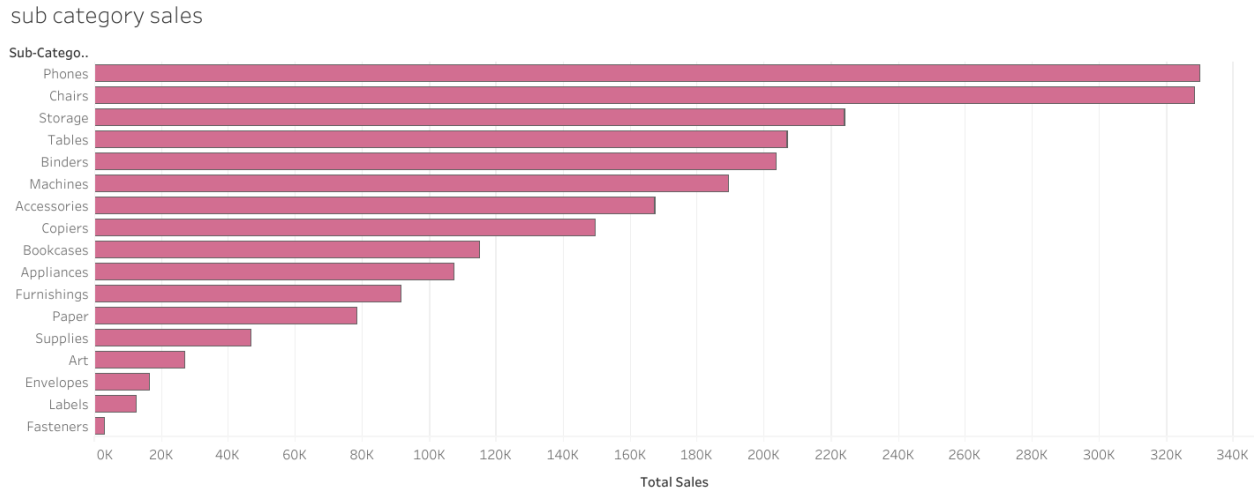
These analyses helped identify patterns in revenue generation and profitability.

8. Data Visualization

An interactive dashboard was created using **Tableau Public** to visually present the insights. Tableau link-
https://public.tableau.com/views/superstoredata_17756424419440/Dashboard1?:language=en-GB&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

1. Sales by Sub-Category

This visualization shows how revenue is distributed across different product sub-categories.



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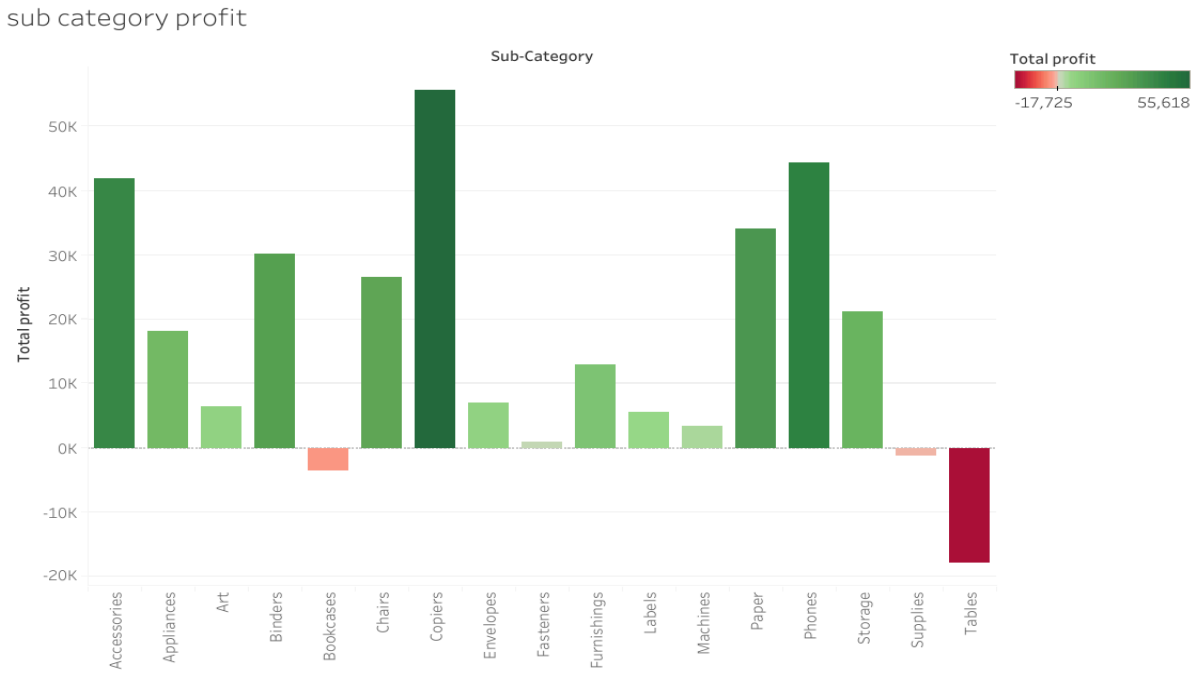
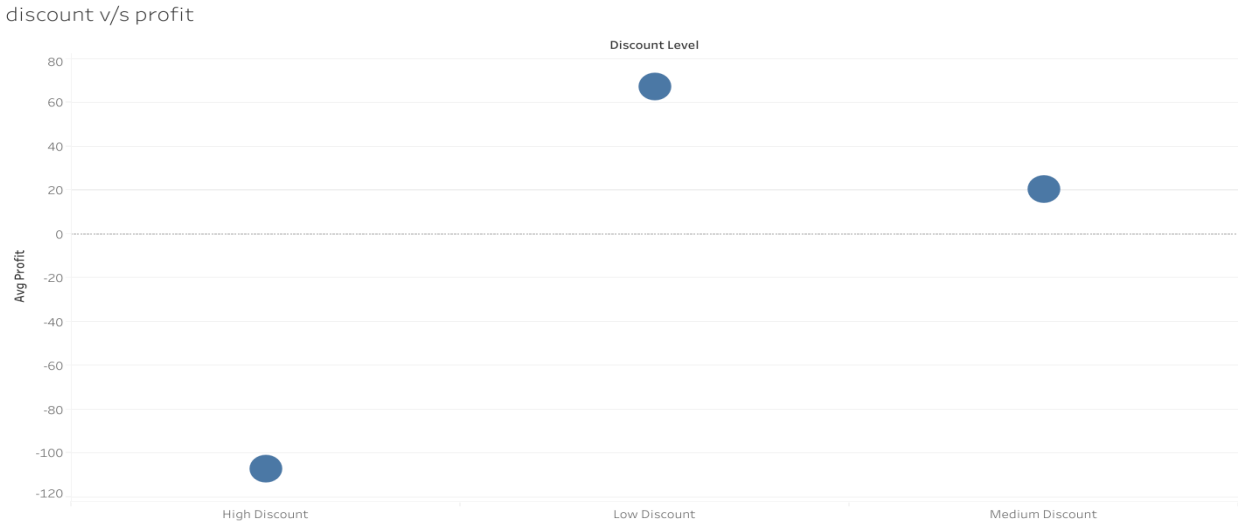
2. Profit by Sub-Category

This chart highlights which product segments contribute positively or negatively to overall profit.



3. Discount vs Profit

A scatter plot visualizing the relationship between discount levels and profitability. This helps identify whether higher discounts lead to reduced profit margins.



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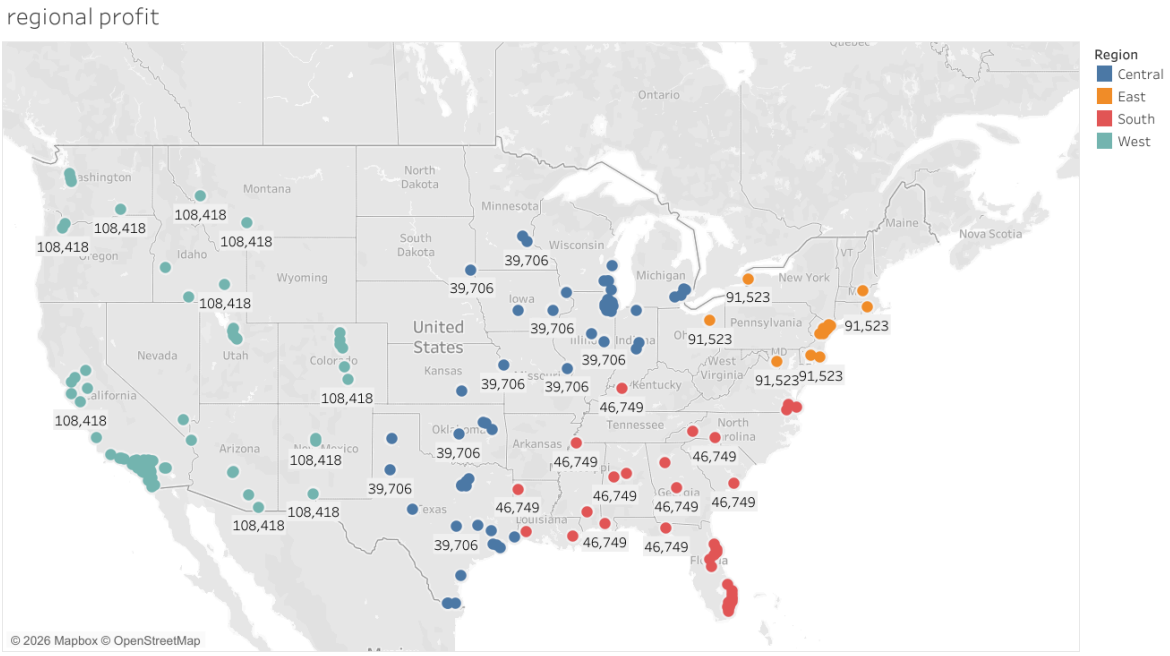
4. Profit by Region

This visualization compares profit performance across different geographic regions.



9. Key Insights

The analysis revealed several important insights:

- Some sub-categories generate high sales but contribute low or negative profit, indicating inefficiencies in pricing or cost structure.
- Higher discount levels are generally associated with lower profitability, suggesting that excessive discounting negatively impacts margins.
- Profitability varies significantly across product segments, highlighting the need for category-level strategy.
- Certain regions outperform others in profit generation, indicating potential opportunities for expansion or optimization.

These insights provide actionable direction for improving business performance.

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10. Recommendations

Based on the analysis, the following recommendations can be made:

- **Optimize discount strategies** to prevent excessive margin loss
- **Focus on high-profit sub-categories** for business growth
- **Re-evaluate low-performing product segments** to improve profitability
- **Leverage high-performing regions** for expansion and targeted marketing

These recommendations can help improve both revenue and profitability.

11. Conclusion

This case study demonstrates how data analytics can be used to uncover valuable business insights from transactional data. By analyzing sales, profit, discount, and regional performance, it is possible to identify key drivers of success and areas for improvement.

The project showcases practical data analytics skills including:

- Data cleaning and preparation
- SQL-based data analysis
- Data visualization and dashboard creation
- Insight generation and business recommendation

The final dashboard provides a clear and interactive way to explore business performance and supports data-driven decision-making.

12. Author

Gopika K
Data Analytics Portfolio Project

Tools Used:
Google Sheets | MySQL | Tableau Public

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